RÉNYI AND TSALLIS ENTROPY MONOTONICITY FOR BERNOULLI SUMS

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Abstract. We prove that the Rényi and Tsallis entropies of a sum of independent Bernoulli random variables are nondecreasing in each success probability on $[0,1/2]^n$ for every order $0 \le q \le 2$, as asserted in the Hillion–Johnson entropy monotonicity conjecture. The proof holds for arbitrary finite n and is entirely real-variable. A quadratic interpolation inequality reduces the sign of the entropy derivative to fair binomial laws. Below order one, the local inequality follows from two integrating-factor identities for an explicit scalar function; above order one, it follows from a local third-derivative estimate. The boundary cases are included, and a two-coin example shows that monotonicity fails at every order $q > 2$. We also obtain a strict quantitative decrease of Hellinger distance to the uniform law on a fixed count alphabet under any genuine coordinate increase.

1. Introduction and main result

For a sum of independent Bernoulli random variables with success probabilities at most $1/2$, does making one coin closer to fair increase the entropy of the sum? Hillion and Johnson proved this for Shannon entropy and formulated the Rényi–Tsallis extension as Conjecture 4.4 in [1]. Their paper also treats order zero, proves the order-two case, and gives counterexamples at every order greater than two.

We prove the full statement on $0 \le q \le 2$ for arbitrary finite Bernoulli sums, also called Poisson–binomial laws. The two open ranges $0 < q < 1$ and $1 < q < 2$ require different local inequalities, but both lead to the same quadratic interpolation on the parameter cube.

For a finite probability mass function $f = (f_k)$ and $q > 0$, $q \ne 1$, the Rényi and Tsallis entropies are

$$H_q(f) = \frac{\log \sum_k f_k^q}{1-q}, \qquad T_q(f) = \frac{1 - \sum_k f_k^q}{q-1}. \tag{1.1}$$

These are the standard families introduced in [2, 3]. All logarithms are natural. At $q = 1$, both are defined as Shannon entropy,

$$H_1(f) = T_1(f) = -\sum_k f_k \log f_k, \qquad 0 \log 0 = 0.$$

At order zero, we set $H_0(f) = \log |\operatorname{supp} f|$ and $T_0(f) = |\operatorname{supp} f| - 1$, without any convention for 0^0.

Theorem 1.1 (Entropy monotonicity)**.** *Let $n \ge 1$. For $p \in [0,1/2]^n$, let f_p be the law of $\sum_{i=1}^n X_i$, where the X_i are independent and $X_i \sim \mathrm{Ber}(p_i)$. If*

$$0 \le p_i \le p_i' \le \tfrac{1}{2} \qquad (1 \le i \le n),$$

then, for every real $0 \le q \le 2$,

$$H_q(f_p) \le H_q(f_{p'}), \qquad T_q(f_p) \le T_q(f_{p'}).$$

The range is sharp: for every real $q > 2$, both inequalities fail for some coordinatewise ordered pair in $[0,1/2]^2$.

The proof first reduces coordinate monotonicity to a sign at a fair moving coin. For a finite log-concave background, we then establish a quadratic lower bound by its two endpoint values.

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Iterating this bound replaces each background coin by either a deterministic zero or a fair coin. Every terminal law is a fair binomial, and symmetry makes its fair-coordinate derivative vanish.

Section 3 proves the scalar estimate for orders below one by elementary differential identities and positive real integrals. Section 4 turns it into the required interpolation inequality. Section 5 proves the same interpolation for orders between one and two. The endpoint orders and sharpness are treated in Section 6, followed by the Hellinger consequence in Section 7.

2. The fair coordinate reduction

2.1. **It suffices to control a fair coordinate.** Fix all coins but one, and call the remaining background law g. Extend every finite sequence by zero to all integer indices. The moving law is

$$f_k(r) = (1-r)g_k + rg_{k-1}, \qquad w_k = g_{k-1} - g_k.$$

For $0 < r < 1$, with identically zero summands omitted,

$$\frac{d^2}{dr^2} T_q(f(r)) = -q \sum_k w_k^2 f_k(r)^{q-2} \le 0. \tag{2.1}$$

The same formula at $q = 1$ is $-\sum_k w_k^2/f_k(r)$. Thus the derivative at $r \le 1/2$ is at least its value at $1/2$. Write that fair-coordinate derivative as

$$\mathscr{D}_q(g) = -\frac{q2^{1-q}}{q-1} \sum_k (g_{k-1} - g_k)(g_{k-1} + g_k)^{q-1}, \qquad q \ne 1, \tag{2.2}$$

and

$$\mathscr{D}_1(g) = -\sum_k (g_{k-1} - g_k) \log(g_{k-1} + g_k). \tag{2.3}$$

Here (2.2) is used only for $q > 0$, $q \ne 1$. The term at a zero pair is zero. These are continuous functionals on finite nonnegative sequences: for $q > 0$, $q \ne 1$, the magnitude of a raw summand is at most $(g_{k-1} + g_k)^q$, and at $q = 1$ use $x|\log x| \to 0$ as $x \downarrow 0$.

It is enough to show $\mathscr{D}_q(g) \ge 0$ for every lower-cube Poisson–binomial background. Rényi and Tsallis have the same derivative sign, since

$$\frac{dH_q}{dr} = \frac{1}{\sum_k f_k^q} \frac{dT_q}{dr}. \tag{2.4}$$

For each fixed positive order, entropy is continuous on the fixed finite alphabet, including zero masses. An interior-coordinate monotonicity inequality therefore extends to the closed moving interval by continuity. No finite derivative at moving parameter zero is required.

2.2. **Log-concavity and the vertices of the parameter cube.** A nonnegative sequence is log-concave if $h_k^2 \ge h_{k-1}h_{k+1}$ and its positive support is an interval. Bernoulli convolution preserves this property. Indeed, if $g_k = ah_k + bh_{k-1}$ with $a, b \ge 0$, then

$$\begin{aligned} g_k^2 - g_{k-1}g_{k+1} &= a^2(h_k^2 - h_{k-1}h_{k+1}) + b^2(h_{k-1}^2 - h_{k-2}h_k) \\ &\quad + ab(h_{k-1}h_k - h_{k-2}h_{k+1}) \ge 0. \end{aligned} \tag{2.5}$$

The mixed determinant is nonnegative by decreasing consecutive ratios on the positive support, and directly at the boundary. Starting with a point mass proves log-concavity for all Bernoulli sums.

At a background parameter vector in $\{0, 1/2\}^m$, deleting the zero coins leaves a fair binomial law. Reflection of its edges preserves $g_{k-1} + g_k$ and negates $g_{k-1} - g_k$. Therefore

$$\mathscr{D}_q(\mathrm{Bin}(j, 1/2)) = 0 \qquad (j \ge 0). \tag{2.6}$$

We will reduce every background to these vertices. The two ranges of q use different local inequalities.

3. A scalar inequality for orders below one

Fix $0 < q < 1$. For $-1 < z < 1$ and $0 \le s \le 1$, define

$$\delta(z) = \left(\frac{1+z}{2}\right)^q - \left(\frac{1-z}{2}\right)^q, \qquad h_q(z) = \frac{z}{\delta(z)}, \quad h_q(0) = \frac{2^{q-1}}{q}, \tag{3.1}$$

$$B_s(z) = (1-sz)^q - 1 - s^2\big[(1-z)^q - 1\big], \qquad K_s(z) = h_q(z)B_s(z). \tag{3.2}$$

The function h_q is smooth and even. We shall prove the following shape property on the whole interval, including its negative half.

Lemma 3.1 (Scalar shape inequality)**.** *For $0 < q < 1$ and $0 \le s \le 1$,*

$$K_s'''(z) \ge 0 \quad (-1 < z < 1), \qquad K_s(0) = K_s''(0) = 0. \tag{3.3}$$

The proof uses an auxiliary function whose positivity is obtained from an integrating factor. Throughout this section, put

$$\rho = 1 - q \in (0,1), \qquad a(x) = 1 - x^2, \qquad c = \rho(2-\rho) = 1 - q^2.$$

Lemma 3.2 (Auxiliary monotonicity)**.** *For $|x| < 1$, let*

$$F(x) = \frac{1}{2}\int_{-1}^{1} (1+tx)^{-\rho}\, dt = \frac{(1+x)^q - (1-x)^q}{2qx}, \qquad F(0) = 1. \tag{3.4}$$

For $0 < x < 1$, define

$$d(x) = \frac{\rho}{a(x)} - \frac{F'(x)}{xF(x)}, \qquad W(x) = a(x)^\rho F(x)^2, \qquad H(x) = a(x)F(x).$$

The function d has a smooth even extension at zero, with $d(0) = c/3$. Moreover,

$$d(x) > 0, \quad d'(x) > 0, \quad H(x) > 0, \quad H'(x) < 0 \qquad (0 < x < 1).$$

Proof. The integral in (3.4) follows by elementary integration. It shows that F is positive, even, and smooth on $(-1,1)$. Indeed, on every compact interval $|x| \le b < 1$, the base $1 + tx$ is bounded below by $1 - b$, so each derivative of the integrand is uniformly bounded for $-1 \le t \le 1$. Differentiation under the integral gives

$$F(0) = 1, \qquad F'(0) = 0, \qquad F''(0) = \frac{\rho(1+\rho)}{3}.$$

The identity

$$\frac{F'(x)}{x} = \int_0^1 F''(tx)\, dt$$

proves smooth removability at zero. Consequently $d(0) = \rho - F''(0) = c/3$ and $d'(0) = 0$.

Consider the differential operator

$$L_q y = ay'' + 2(q-1)xy' - q(q-1)y.$$

Direct differentiation gives, whenever $1 + bx > 0$,

$$L_q[(1+bx)^q] = q(q-1)(b^2-1)(1+bx)^{q-2}. \tag{3.5}$$

In particular $L_q[xF] = 0$. Expanding this identity yields

$$aF'' + \left[\frac{2}{x} - 2(1+\rho)x\right]F' - \rho(1+\rho)F = 0 \qquad (x \ne 0). \tag{3.6}$$

Set $u = F'/F = x(\rho/a - d)$. Dividing (3.6) by F and substituting this expression gives

$$0 = a(u' + u^2) + [2/x - 2(1+\rho)x]u - \rho(1+\rho) = \frac{c}{a} - 3ad - axd' + ax^2d^2.$$

Thus

$$xd' + 3d = x^2d^2 + \frac{c}{a^2}. \tag{3.7}$$

Differentiating the equality (3.7), rather than a sign inequality, gives

$$d'' + \left(\frac{4}{x} - 2xd\right) d' = 2d^2 + \frac{4c}{a^3}.$$

The definition of W implies $W'/W = -2xd$, and hence

$$(x^4 W d')' = x^4 W \left(2d^2 + \frac{4c}{a^3}\right). \tag{3.8}$$

Since $d'(x) = O(x)$ and $W(0) = 1$, the left-hand primitive tends to zero at the origin. The right-hand side is $O(x^4)$ there. Integrating (3.8) from zero therefore gives the exact formula

$$d'(x) = \frac{\displaystyle\int_0^x t^4 W(t) \left(2d(t)^2 + \frac{4c}{(1-t^2)^3}\right) dt}{x^4 W(x)} > 0. \tag{3.9}$$

No sign of d was needed to obtain this inequality: the only such term in the integrand is a square. Since $d(0) = c/3 > 0$, we also have $d > 0$. Finally,

$$\frac{H'}{H} = -x \left(\frac{2-\rho}{a} + d\right) < 0, \tag{3.10}$$

while $H > 0$ follows directly from $a, F > 0$. □

Proof of Lemma 3.1. The cases $s = 0, 1$ give $K_s \equiv 0$, so assume $0 < s < 1$. By (3.4),

$$\widehat{K}_s(z) := 2^{1-q} K_s(z) = \frac{B_s(z)}{qF(z)}.$$

Because F is even and $B_s(0) = B_s''(0) = 0$,

$$\widehat{K}_s(0) = \widehat{K}_s''(0) = 0, \qquad \widehat{K}_s'(0) = -s(1-s).$$

For $e \in \{-1, 1\}$ and $0 \le x < 1$, set

$$Z_e(x) = e\widehat{K}_s(ex).$$

The sign convention ensures that

$$Z_e'''(x) = \widehat{K}_s'''(ex), \qquad Z_e(0) = Z_e''(0) = 0. \tag{3.11}$$

We now derive the same positive curvature equation for both choices of e.

Equation (3.5), applied to every term of B_s, gives

$$L_q[B_s(ex)] = q(q-1)(1-s^2)[1 - (1-esx)^{q-2}]. \tag{3.12}$$

For any smooth Z, equations (3.6) and $F'/F = x(\rho/a - d)$ give

$$L_q[FZ] = aF\{Z'' - 2d(xZ' - Z)\}. \tag{3.13}$$

For completeness, the coefficients of Z' and Z on the left are $2aF' + 2(q-1)xF = -2aFxd$ and $aF'' + 2(q-1)xF' - q(q-1)F = 2aFd$, respectively. Since $FZ_e = eB_s(ex)/q$, it follows that

$$Z_e'' - 2d(xZ_e' - Z_e) = f_e, \qquad f_e = \frac{\rho(1-s^2)}{H} N_e, \tag{3.14}$$

where

$$N_e(x) = e[(1-esx)^{-1-\rho} - 1], \qquad N_e'(x) = (1+\rho)s(1-esx)^{-2-\rho} \ge 0.$$

All bases are positive. Since $N_e(0) = 0$ and $H' \le 0$, we have

$$f_e \ge 0, \qquad f_e' = \rho(1-s^2)\left(\frac{N_e'}{H} - \frac{N_e H'}{H^2}\right) \ge 0. \tag{3.15}$$

Put $g_e = Z_e''$ and $V_e = xZ_e' - Z_e$. By (3.11),

$$V_e(0) = 0, \qquad V_e' = xg_e, \qquad g_e = f_e + 2dV_e.$$

Since $W'/W = -2xd$, these identities imply

$$(WV_e)' = xWf_e.$$

Integration, with $W(0) = 1$, gives a positive real integral:

$$V_e(x) = \frac{1}{W(x)} \int_0^x tW(t) f_e(t)\, dt \geq 0. \tag{3.16}$$

Differentiating the exact identity $g_e = f_e + 2dV_e$ now yields

$$g_e' = f_e' + 2d'V_e + 2dV_e' = f_e' + 2xdf_e + (2d' + 4xd^2)V_e \geq 0. \tag{3.17}$$

Each term has a separately established nonnegative sign, by Lemma 3.2, (3.15), and (3.16). Thus (3.11) proves the third derivative inequality on both halves of $(-1, 1)$. Smoothness covers the origin. Multiplication by $2^{q-1} > 0$ returns to K_s and proves the lemma. □

All integrations above take place on compact subintervals of $[0, 1)$. No limit of the auxiliary differential equations at $x = 1$ is needed. The values at $z = \pm 1$ required below will instead be handled by continuity of homogeneous expressions.

4. Quadratic interpolation for orders below one

For positive a, b, define the symmetric, degree-zero function

$$\mathscr{H}_q(a, b) = h_q\left(\frac{a-b}{a+b}\right) = \frac{(a-b)(a+b)^{q-1}}{a^q - b^q}, \qquad \mathscr{H}_q(a, a) = \frac{2^{q-1}}{q}. \tag{4.1}$$

For a positive log-concave triple (v, u, t), let

$$A = pv + (1-p)u, \quad C = pu + (1-p)t, \quad x = A + C, \quad w = A - C, \tag{4.2}$$

and put

$$J(p) = -\mathscr{H}_q(v, u)A^q + \mathscr{H}_q(u, t)C^q + wx^{q-1}. \tag{4.3}$$

Proposition 4.1 (Local quadratic interpolation)**.** *If $0 < q < 1$, $u^2 \geq vt$, and $0 \leq p \leq 1/2$, then*

$$J(p) \geq (1-2p)^2 J(0) + 4p(1-p)J(1/2). \tag{4.4}$$

Also $J(0) = J(1) = u^q[\mathscr{H}_q(u, t) - \mathscr{H}_q(v, u)]$.

Proof. The endpoint identity follows from (4.1). For the inequality, normalize by

$$L = v + 2u + t, \quad z = \frac{v-t}{L}, \quad \kappa = \frac{u^2 - vt}{L^2}, \quad \theta = \frac{1+z}{2}, \quad \eta = \frac{1-z}{2}, \quad s = 1 - 2p.$$

The two edge scores are

$$\alpha = z - \frac{2\kappa}{\theta}, \qquad \gamma = z + \frac{2\kappa}{\eta}.$$

Indeed,

$$\frac{v}{L} = \theta^2 - \kappa, \quad \frac{u}{L} = \theta\eta + \kappa, \quad \frac{t}{L} = \eta^2 - \kappa,$$

and $\alpha = (v-u)/(v+u)$, $\gamma = (u-t)/(u+t)$. In particular the entire interval from 0 to the given κ is feasible at fixed L, z. Moreover,

$$A = \frac{L}{2}\theta(1 - s\alpha), \qquad C = \frac{L}{2}\eta(1 - s\gamma),$$
$$x = \frac{L}{2}(1 - sz), \qquad w = \frac{L}{2}[z(1 - sz) + 4s\kappa].$$

Direct substitution shows that the gap in (4.4), divided by $(L/2)^q$, is

$$\begin{aligned} G(\kappa) = & -\theta^q K_s(\alpha) + \eta^q K_s(\gamma) + zB_s(z) \\ & + 4\kappa\big[s(1 - sz)^{q-1} - s^2(1 - z)^{q-1}\big]. \end{aligned} \tag{4.5}$$

At $\kappa = 0$, both scores are z, and $h_q(z)(\theta^q - \eta^q) = z$, so $G(0) = 0$. Differentiating at fixed z gives

$$G'(\kappa) = 2\big[\theta^{q-1}K_s'(\alpha) + \eta^{q-1}K_s'(\gamma)\big] - \frac{4}{q}B_s'(z). \tag{4.6}$$

Let $S = \theta^{q-1} + \eta^{q-1}$. The weighted mean of the scores is

$$m = z + \frac{2\kappa}{S}(\eta^{q-2} - \theta^{q-2}).$$

It moves away from zero: $m \geq z \geq 0$ if $z \geq 0$, and $m \leq z \leq 0$ if $z \leq 0$. Lemma 3.1 says K_s' is convex, and $K_s''(0) = 0$ implies K_s' increases on the positive half and decreases on the negative half. Jensen's inequality therefore gives

$$\theta^{q-1}K_s'(\alpha) + \eta^{q-1}K_s'(\gamma) \geq SK_s'(m) \geq SK_s'(z). \tag{4.7}$$

To determine the sign of the resulting lower bound, note that $\delta' = qS/2$ and $B_s = \delta K_s/z$. Consequently

$$2SK_s'(z) - \frac{4}{q}B_s'(z) = \frac{4}{q}[z\delta'(z) - \delta(z)]\left(\frac{K_s(z)}{z}\right)'. \tag{4.8}$$

No power series is needed here: direct differentiation gives

$$\delta''(z) = 2^{-q}q(q-1)[(1+z)^{q-2} - (1-z)^{q-2}],$$

which has the sign of z. Thus $M(z) = z\delta'(z) - \delta(z)$ satisfies $M(0) = 0$ and $M'(z) = z\delta''(z) \geq 0$, so $M(z)$ also has the sign of z. Meanwhile $zK_s' - K_s$ vanishes at zero and has derivative $zK_s''(z) \geq 0$ on both sides of zero. It therefore also has the sign of z, as does $(K_s/z)'$. The product in (4.8) is nonnegative. At zero the quotient is smoothly removable, since

$$\frac{K_s(z)}{z} = \int_0^1 K_s'(tz)\,dt, \qquad \left(\frac{K_s(z)}{z}\right)'\Bigg|_{z=0} = \frac{1}{2}K_s''(0) = 0.$$

Equations (4.6)–(4.8) show $G'(\kappa) \geq 0$. Integrating from 0 to κ proves the proposition. □

4.1. **Zeros and exact telescoping.** The function h_q is bounded and continuous on $[-1, 1]$, with $h_q(\pm 1) = 1$. Hence

$$\mathscr{H}_q(a, b)[pa + (1-p)b]^q$$

extends continuously to all nonnegative pairs, with value zero at $(0, 0)$. The raw term wx^{q-1} extends as well, since its magnitude is at most x^q. Every nonnegative log-concave triple can be approximated by positive ones: take $v_\varepsilon = v + \varepsilon$, $t_\varepsilon = t + \varepsilon$, and $u_\varepsilon = \max\{u, \sqrt{v_\varepsilon t_\varepsilon}\} + \varepsilon$. Thus Proposition 4.1 includes zero entries by continuity.

Now let h be any finite log-concave sequence and $g_k(p) = (1-p)h_k + ph_{k-1}$. At the triple $(v, u, t) = (h_{k-2}, h_{k-1}, h_k)$, the masses in (4.2) are $A = g_{k-1}(p)$ and $C = g_k(p)$. The first two terms of (4.3) cancel exactly when summed over all k. All terms outside the finite support are zero. Multiplying the summed inequality by $q2^{1-q}/(1-q) > 0$ proves the following.

Theorem 4.2 (Below-one Bellman inequality)**.** *For $0 < q < 1$, a finite log-concave h, and $0 \leq p \leq 1/2$,*

$$\mathscr{D}_q(h * \mathrm{Ber}(p)) \geq (1-2p)^2\mathscr{D}_q(h) + 4p(1-p)\mathscr{D}_q(h * \mathrm{Ber}(1/2)). \tag{4.9}$$

Both coefficients in (4.9) are nonnegative and sum to one. For a Poisson–binomial background with parameters $p_1, \ldots, p_m \in [0, 1/2]$, apply the inequality successively to each parameter. The remaining background at each step is log-concave by (2.5). The resulting lower bound is a convex combination of the values at all vertices $\varepsilon \in \{0, 1\}^m$, with weights

$$W_\varepsilon = \prod_{\varepsilon_i=0}(1-2p_i)^2 \prod_{\varepsilon_i=1} 4p_i(1-p_i).$$

Each vertex value is zero by (2.6), so $\mathscr{D}_q(g) \geq 0$. Equations (2.1) and (2.4) prove Theorem 1.1 for $0 < q < 1$.

5. Quadratic interpolation for orders between one and two

We give a separate proof for $1 < q < 2$. Fix $0 < p < 1$ and a positive log-concave triple (v, u, t). Use A, C, x, w from (4.2), and set

$$B = v - t, \qquad E = v - 2u + t, \qquad T = u^2 - vt \geq 0.$$

Define

$$\begin{aligned} \mathcal{K}_{p,q}(v,u,t) &= 3EB^2x^{q-3} + (q-3)wB^3x^{q-4}, \\ \mathcal{H}_{p,q}(a,b) &= 2^{q-1}\frac{(a-b)^3}{[pa+(1-p)b]^{3-q}}, \\ R_q &= \mathcal{K}_{p,q}(v,u,t) - \mathcal{H}_{p,q}(v,u) + \mathcal{H}_{p,q}(u,t). \end{aligned} \tag{5.1}$$

Lemma 5.1 (Upper-range local inequality)**.** *$R_q \geq 0$ for every positive log-concave triple and $1 \leq q \leq 2$. The inequality extends to nonnegative log-concave triples by continuity.*

Proof. The elementary identities

$$wB - Ex = 2T, \qquad (v-u)x = AB - T, \qquad (u-t)x = CB + T \tag{5.2}$$

give two exact endpoint formulas:

$$R_1 = \frac{T^2\{3wB\,AC + (A^2+C^2)T\}}{x^3A^2C^2}, \qquad R_2 = \frac{2T^3}{x^2AC}. \tag{5.3}$$

They follow by substituting (5.2) into the two cubic correction terms and expanding. For example, at $q = 1$, multiply by $x^3A^2C^2$: the resulting identity is

$$3EB^2xA^2C^2 - 2wB^3A^2C^2 - (AB-T)^3C^2 + (CB+T)^3A^2 = T^2[3wBAC + (A^2+C^2)T],$$

using $Ex = wB - 2T$, $x = A + C$ and $w = A - C$. At $q = 2$, multiplication by x^2AC similarly gives

$$3EB^2xAC - wB^3AC - 2(AB-T)^3C/x + 2(CB+T)^3A/x = 2T^3.$$

The formula for R_2 is nonnegative. The formula for R_1 is immediate when $wB \geq 0$. Otherwise the reflection $(v, u, t, p) \mapsto (t, u, v, 1-p)$ lets us suppose $B > 0$ and $w < 0$. Log-concavity and $w = p(v-u) + (1-p)(u-t)$ then force $u > v > t$. Put $a = u - v$, $b = u - t$. Here $0 < a < b \leq u$, $B = b - a$, and $-w \leq a$. Hence

$$T = u(a+b) - ab \geq b^2 \geq 4a(b-a) \geq 4(-wB).$$

The first difference is $(u-b)(a+b) \geq 0$ and the second inequality is $(b-2a)^2 \geq 0$. Writing $d = -wB > 0$, the bracket in R_1 is at least

$$d[4(A^2+C^2) - 3AC] \geq 5dAC > 0.$$

This proves the $q = 1$ endpoint.

For $1 < q < 2$, first suppose $E \leq 0$. For differentiable ψ, define the linear functional

$$\mathcal{L}(\psi) = 3EB^2\psi(x) + wB^3\psi'(x) - 4[(v-u)^3\psi(2A) - (u-t)^3\psi(2C)]. \tag{5.4}$$

It equals R_q when $\psi(y) = y^{q-3}$. The positive integral representation

$$y^{q-3} = \frac{1}{I_q}\int_0^\infty r^{q-2}(y+r)^{-2}\,dr, \quad I_q = \int_0^\infty \frac{r^{q-2}}{(1+r)^2}\,dr > 0 \tag{5.5}$$

shows that R_q is a positive weighted integral of R_1 evaluated on the common-shifted triple $(v + r/2, u + r/2, t + r/2)$. Indeed, the differences B, E, w remain fixed, while $x, 2A, 2C$ all increase by r. The shifted determinant is $T - rE/2 \geq 0$, so every such R_1 is nonnegative. Differentiation and the integral interchange are justified locally uniformly for positive arguments by integrable dominating powers: near zero the weight is r^{q-2}, and at infinity the integrands are $O(r^{q-4})$ or faster; differentiating a shifted inverse square only improves its decay. Thus $R_q \geq 0$ if $E \leq 0$.

The remaining case can be covered whenever $wB \geq 0$. Normalize by

$$z = w/x, \qquad r = B/x, \qquad \tau = T/x^2.$$

Reflection lets us assume $z, r \geq 0$; $|z| < 1$ and $\tau \geq 0$. Define

$$\begin{aligned} U(z) &= 2 - (1+z)^{q-1} - (1-z)^{q-1}, \\ V(z) &= qz - \tfrac{1}{2}[(1+z)^q - (1-z)^q], \\ W(z) &= (1+z)^{q-3} + (1-z)^{q-3}. \end{aligned}$$

Expanding the two cubes using (5.2) gives exactly

$$\frac{R_q}{x^q} = V(z)r^3 - 3U(z)r^2\tau + \frac{6U'(z)}{q-1}r\tau^2 + 4W(z)\tau^3. \tag{5.6}$$

For $0 < z < 1$, all four functions U, U', V, W are positive, and

$$U'' = (q-1)(2-q)W > 0, \qquad V(z) = \frac{q}{2}\int_0^z U(a)\,da.$$

Since U' is increasing,

$$U(z)^2 = 2\int_0^z U(a)U'(a)\,da \leq \frac{4}{q}U'(z)V(z). \tag{5.7}$$

For $r > 0$, the first three terms in (5.6) form a quadratic in τ with positive constant and quadratic coefficients. Its squared linear coefficient divided by four times their product is at most

$$\frac{9U^2r^4}{4Vr^3\,[6U'r/(q-1)]} \leq \frac{3(q-1)}{2q} < 1.$$

Thus the quadratic is nonnegative, and so is the last term in (5.6). If $z = 0$ or $r = 0$, the first three terms vanish and the conclusion is immediate.

These cases exhaust all triples: the identity $Ex = wB - 2T$ shows that $wB < 0$ implies $E < 0$. This proves $R_q \geq 0$. For fixed $p \in (0,1)$, both the two-point correction and the local kernel have magnitude bounded by a constant times the qth power of the sum of their arguments. They therefore extend continuously at zero. Approximation by positive log-concave triples, as in Section 4, finishes the proof. □

5.1. **The same quadratic Bellman inequality above one.** Let h be a finite log-concave background and $g_k(p) = (1-p)h_k + ph_{k-1}$. Write $D(p) = \mathscr{D}_q(g(p))$. For the local triple at index k, $x' = B$ and $w' = E$. Differentiating (2.2) three times yields

$$D'''(p) = q(2-q)2^{1-q}\sum_k \mathcal{K}_{p,q}(h_{k-2}, h_{k-1}, h_k) \geq 0. \tag{5.8}$$

The inequality follows from Lemma 5.1, because all two-point correction terms telescope. The function D is continuous on $[0,1]$ and C^3 on $(0,1)$: every non-identically-zero adjacent mass sum is positive there. Also $D(0) = D(1)$ by shift invariance. No derivative regularity at the endpoints is needed.

Lemma 5.2 (Three-node quadratic interpolation)**.** *If $D \in C[0,1] \cap C^3(0,1)$, $D''' \geq 0$, and $D(0) = D(1)$, then*

$$D(p) \geq (1-2p)^2 D(0) + 4p(1-p)D(1/2), \qquad 0 \leq p \leq 1/2.$$

Proof. Let $Q(p) = (1-2p)^2D(0) + 4p(1-p)D(1/2)$, the quadratic interpolant at $0, 1/2, 1$. For $0 < p < 1/2$, subtract from $D - Q$ the cubic $\lambda t(t-1/2)(t-1)$ chosen to make the difference zero at $t = p$. This difference has four distinct zeros in $[0,1]$. Three applications of Rolle's theorem give $6\lambda = D'''(\xi) \geq 0$ for some $\xi \in (0,1)$. Since $p(p-1/2)(p-1) > 0$, $D(p) - Q(p) \geq 0$. At $p = 0, 1/2$ there is equality by definition. □

Consequently the below-one Bellman inequality (4.9) holds also for every $1 < q < 2$, with exactly the same coefficients, for every finite log-concave background. The same product-weight rounding argument already given after Theorem 4.2 therefore applies without modification: iterating over the finite background parameters gives a convex combination of fair-binomial vertex values, all zero. Thus $\mathscr{D}_q(g) \geq 0$, proving Theorem 1.1 for $1 < q < 2$ by the common reduction.

6. Endpoint orders and sharpness

At $q = 0$, let r be the number of strictly positive Bernoulli parameters. On the lower cube the sum has support $\{0, \dots, r\}$, so $T_0 = r$ and $H_0 = \log(r + 1)$. Increasing a coordinate cannot decrease r, proving this endpoint directly.

At $q = 1$, take the pointwise limit $T_q(f) \to -\sum_k f_k \log f_k$ on each fixed finite support. Passing to the limit in the already proved monotonicity inequality gives Shannon monotonicity, including all zero-probability boundary cases.

At $q = 2$, the fair derivative is identically zero for every finite background:

$$\mathscr{D}_2(g) = -\sum_k (g_{k-1}^2 - g_k^2) = 0.$$

Coordinate concavity again gives monotonicity for the moving parameter below one half. This completes the positive part of Theorem 1.1.

6.1. A counterexample for every $q > 2$.

Fix $q > 2$ and set

$$r = \frac{q-2}{q(q-1)} \in (0, 1/2).$$

The upper bound follows from $q(q-1) - 2(q-2) = q^2 - 3q + 4 = (q - 3/2)^2 + 7/4 > 0$. For the one-coin background $g = (1 - r, r)$, formula (2.2) becomes

$$\mathscr{D}_q(g) = -\frac{q 2^{1-q}}{q-1}[1 - 2r - (1-r)^q + r^q].$$

The third derivative of $(1 - r)^q$ in r is negative. Taylor's theorem therefore gives

$$(1-r)^q \leq 1 - qr + \frac{q(q-1)}{2} r^2.$$

The bracket is consequently at least

$$(q-2)r - \frac{q(q-1)}{2} r^2 + r^q = \frac{q-2}{2} r + r^q > 0.$$

Thus $\mathscr{D}_q(g) < 0$. Continuity places a negative derivative at moving parameters strictly below and sufficiently close to $1/2$ as well. Rényi has the same sign by (2.4). For example, $q = 3$ and $r = 1/6$ give exactly $\mathscr{D}_3(g) = -5/144$.

7. A quantitative Hellinger consequence

The order $q = 1/2$ gives a quantitative comparison in Hellinger distance. The reference distribution is uniform on a fixed alphabet, even when some Bernoulli parameters vanish.

Corollary 7.1 (Strict quantitative improvement toward fixed-alphabet uniform)**.** *Let $n \geq 1$, let f be the law of a sum of independent Bernoulli variables with parameters in $[0, 1/2]^n$, and let U_n be uniform on $\{0, \dots, n\}$. Use the convention*

$$h^2(f, U_n) = 1 - \sum_{k=0}^{n} \sqrt{\frac{f_k}{n+1}} = \frac{1}{2} \sum_{k=0}^{n} \left(\sqrt{f_k} - \frac{1}{\sqrt{n+1}} \right)^2.$$

Every genuine coordinate increase strictly decreases this distance. More precisely, freeze the other $n-1$ coins with law g, extend it by zero to all integers, put $w_k = g_{k-1} - g_k$, and write $f_k(r) = (1-r)g_k + rg_{k-1}$. For $0 \le a < b \le 1/2$,

$$h^2(f(a), U_n) - h^2(f(b), U_n) \ge \frac{(1/2-a)^2 - (1/2-b)^2}{8\sqrt{n+1}} \sum_k w_k^2 > 0. \tag{7.1}$$

Consequently a coordinatewise ordered pair of distinct parameter vectors gives strict decrease. Independent coordinate thinning $p_i \mapsto t_i p_i$, $0 \le t_i \le 1$, cannot decrease the distance, and increases it strictly if at least one parameter changes.

Proof. At order $1/2$,

$$T_{1/2}(f) = 2\left(\sum_k \sqrt{f_k} - 1\right), \qquad h^2(f(a), U_n) - h^2(f(b), U_n) = \frac{T_{1/2}(f(b)) - T_{1/2}(f(a))}{2\sqrt{n+1}}.$$

For $0 < r < 1/2$, every non-identically-zero $f_k(r)$ is positive and at most one. Write $S_g = \sum_k w_k^2$. The exact second derivative (2.1) gives

$$\frac{d^2}{dr^2} T_{1/2}(f(r)) = -\frac{1}{2} \sum_k w_k^2 f_k(r)^{-3/2} \le -\frac{1}{2} S_g.$$

Theorem 1.1 at the open order $1/2$, through its established fair-derivative sign, gives

$$\frac{d}{dr} T_{1/2}(f(r))\Big|_{r=1/2} = \mathscr{D}_{1/2}(g) \ge 0.$$

Backward integration yields

$$\frac{d}{dr} T_{1/2}(f(r)) \ge \frac{1}{2}(1/2 - r) S_g.$$

Integrating from a to b yields

$$T_{1/2}(f(b)) - T_{1/2}(f(a)) \ge \frac{1}{4}[(1/2-a)^2 - (1/2-b)^2] S_g.$$

Division by $2\sqrt{n+1}$ proves (7.1) for $a > 0$. If $a = 0$, first integrate from $\varepsilon > 0$ and pass to $\varepsilon \downarrow 0$ by finite-alphabet continuity; no finite derivative at zero is used. A finite nonzero probability sequence cannot satisfy $g_{k-1} = g_k$ for every integer, so $S_g > 0$. The square difference is positive for $a < b \le 1/2$, also when $b = 1/2$. A finite path of coordinate changes proves the vector comparison and its thinning consequence. The reference U_n never changes along this path. □

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